

The Virial Partition from Atoms to the Horizon

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<https://github.com/hmahaffeyges/IAM-Validation>

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Abstract

For any system bound by a potential that falls as $1/r$, the virial theorem fixes the time-averaged kinetic energy at exactly half the magnitude of the potential energy. We follow this one half through every scale at which it can be checked: the hydrogen atom, atoms and molecules at the Hartree–Fock limit, the Sun, white dwarfs, clusters of galaxies, the horizon of a black hole, and the horizon of the universe. We state the thermodynamic content of the kinetic half: when a $1/r$ system binds from rest, the energy it releases equals its kinetic half, and Landauer’s principle makes that the minimum cost of the irreversible binding. For a Schwarzschild black hole the Bekenstein–Hawking entropy counted in bits, priced at $k_B T_H \ln 2$ each, equals exactly $\frac{1}{2} M c^2$ (the Smarr relation). At the cosmic horizon the same half fixes the coupling $\beta_m = \Omega_m/2$ of the Informational Actualization Model, which enters every Planck chain at its predicted value and is consistent with Planck 2018 ($\Delta\chi^2 = +0.54$ against Λ CDM, no added parameter). Every number is recomputed by the script `docs/verification/scripts/verify_virial_atoms_to_horizon.py`.

1 The theorem

For a bound system with potential energy homogeneous of degree k in the coordinates, the time average of $d(\sum_i \mathbf{r}_i \cdot \mathbf{p}_i)/dt$ vanishes, and Euler’s theorem gives $2\langle K \rangle = k\langle V \rangle$. Gravity and the Coulomb interaction have $k = -1$:

$$2\langle K \rangle + \langle V \rangle = 0, \quad \langle K \rangle = \frac{1}{2}|\langle V \rangle|, \quad E = \langle K \rangle + \langle V \rangle = -\langle K \rangle. \quad (1)$$

The degree is fixed by geometry: in three spatial dimensions Gauss’s law makes the field of a point source fall as $1/r^2$, so the potential falls as $1/r$. The half does not depend on the number of bodies, their masses or the scale.

2 Atoms and molecules

Hydrogen. The ground state has $E_1 = -13.6057$ eV, $\langle K \rangle = 13.6057$ eV and $\langle V \rangle = -27.2114$ eV: $\langle K \rangle/|\langle V \rangle| = 1/2$. An electron captured from rest into this state emits a photon of 13.6057 eV, equal to its kinetic half.

Many-electron atoms and molecules. For the 20 atoms H–Xe and 10 molecules (H_2 to CO_2) at the Hartree–Fock limit (Clementi & Roetti 1974; Bunge et al. 1993), $T/|V| = 1/2$ to the precision of the tabulated energies. For a converged variational wavefunction this is exact: optimising under a uniform scaling of the coordinates enforces the quantum virial theorem. These systems show that quantum mechanics obeys Eq. (1) with the electrostatic interaction; they are not an independent measurement of it.

3 Stars

The Sun. For a star in hydrostatic equilibrium the theorem gives $2K_{\text{th}} + U = 0$. A contracting star radiates half the gravitational energy it releases and stores the other half as heat. For a polytrope of index 3, $U = -\frac{3}{2}GM^2/R = -5.69 \times 10^{41} \text{ J}$ for the Sun; contraction alone could power the present luminosity for $|U|/2L = 24 \text{ Myr}$. The Sun is 4.6 Gyr old, so it must burn hydrogen — the argument of Kelvin and Helmholtz, settled by the virial theorem.

White dwarfs. In a white dwarf the degenerate electrons become relativistic as the mass grows; the kinetic energy then scales as $1/R$ like the gravitational energy, and Eq. (1) has a solution only at one mass,

$$M_{\text{Ch}} = \frac{\omega_3^0 \sqrt{3\pi}}{2} \left(\frac{\hbar c}{G} \right)^{3/2} \frac{1}{(\mu_e m_u)^2} = 1.456 M_{\odot} \quad (\mu_e = 2), \quad (2)$$

with $\omega_3^0 = 2.01824$ (Chandrasekhar 1931, 1939). No white dwarf has been found above it; type Ia supernovae mark carbon–oxygen white dwarfs that reach it.

4 Clusters of galaxies

Zwicky (1933) applied Eq. (1) to the velocities of the galaxies in the Coma cluster and found far more mass than the light suggests. Virial (dynamical) masses of clusters are now compared with weak-lensing and X-ray masses in large samples; they agree at the tens-of-percent level, set by projection, departures from equilibrium in merging systems and non-thermal support.

Simulated dark-matter halos. Cosmological N -body simulations measure the virial ratio of every halo directly. Within the virial radius the published values are $2T/|U| \approx 1.1$ – 1.3 , rising slowly with mass (Bett et al. 2007; Neto et al. 2007, “median slightly greater than unity”; Power, Knebe & Knollmann 2012, $\eta = 2T/|W| \approx 1.15$ at 10^{12} to 1.25 at $10^{15} h^{-1} M_{\odot}$; Klypin et al. 2016). The excess over 1 is the pressure of matter still falling through the boundary: with the surface-pressure term included, Klypin et al. find $2K/|W| \approx 1.02$ – 1.17 . Halos are called relaxed when $2T/|U| < 1.35$ (Neto et al. 2007). Collisionless halos, assembled without radiating, sit at the virial equilibrium to within the infall still under way.

5 What the kinetic half is

Let a $1/r$ system fall from rest at infinity ($E_i = 0$) into a stable bound state $E_f < 0$ and release the difference.

1. First law: the energy released is $Q = -E_f$.
2. Eq. (1): $-E_f = \langle K \rangle$, so $Q = \langle K \rangle$.
3. Second law: the transition is irreversible, producing entropy $\Delta S \geq Q/T$ in surroundings at temperature T .
4. Landauer (1961): the minimum cost of producing ΔS is $T\Delta S$; the bound is met when $\Delta S = Q/T$.

$$\langle K \rangle = Q = T\Delta S_{\text{min}} = E_{\text{Landauer}} = \frac{1}{2}|\langle V \rangle|. \quad (3)$$

Step 2 is the mechanical theorem, so this does not derive Eq. (1); it identifies its kinetic half with the minimum thermodynamic cost of forming the bound state. It applies wherever the binding energy is released — the photon of atomic recombination, the radiation of a contracting cloud or star. A collisionless system (a dark-matter halo) relaxes without radiating; there the half describes the redistribution of energy in violent relaxation, and the thermodynamic reading

concerns the information written in that relaxation. Jacobson (1995) obtained the Einstein equations from $\delta Q = T dS$ on local horizons; the identity above is the same relation applied at the boundary of a forming bound state.

6 The black-hole horizon

A Schwarzschild black hole of mass M has Bekenstein–Hawking entropy and Hawking temperature

$$S = \frac{k_B c^3 A}{4G\hbar} = \frac{4\pi G k_B M^2}{\hbar c}, \quad T_H = \frac{\hbar c^3}{8\pi G k_B M}. \quad (4)$$

Counting the entropy in bits, $N = S/(k_B \ln 2)$, and pricing each bit at Landauer’s $k_B T_H \ln 2$,

$$N k_B T_H \ln 2 = T_H S = \frac{1}{2} M c^2. \quad (5)$$

For one solar mass, $N = 1.51 \times 10^{77}$ bits at $T_H = 6.17 \times 10^{-8}$ K; for the black holes in the Galactic centre ($4.3 \times 10^6 M_\odot$) and in M87 ($6.5 \times 10^9 M_\odot$) the ratio is the same, 0.5000000000. This is the Smarr relation $M c^2 = 2 T_H S$ of general relativity (Smarr 1973; Bardeen, Carter & Hawking 1973). Read with Landauer’s principle, half of a black hole’s rest energy is the cost of the information on its horizon — the same half as in Eq. (1). For a rotating (Kerr) black hole, $M c^2 = 2 T_H S + 2 \Omega_H J$, and the horizon share is $T_H S / M c^2 = \sqrt{1 - \chi^2}/2$: 0.433 at $\chi = 0.5$, 0.218 at $\chi = 0.9$, 0.032 at $\chi = 0.998$; the rest is rotational energy.

7 The cosmic horizon

The Informational Actualization Model (IAM) adds to the horizon first law of Jacobson and of Cai & Kim (2005) the entropy of the classical records produced as matter decoheres gravitationally, $-dE = T_H d(S_{\text{geo}} + S_{\text{info}})$. The share of the matter density that enters the record term is the kinetic half of the binding energy of collapsed structure:

$$\beta_m = \frac{\Omega_m}{2} = 0.15765 \quad (\Omega_m = 0.3153, \text{ Planck 2018}). \quad (6)$$

The coupling is a prediction. It enters every Planck chain at this value and is never sampled. In the three modified-CAMB chains against the Planck likelihood, $\Delta\chi^2 = +0.54$ against Λ CDM with the same number of parameters, σ_8 moves from 0.809 to 0.800, and the standard parameters are unchanged within their errors. Its decisive test is the growth of structure measured by Euclid: the coupling gives $\mu(z=0) = 0.864$ ($\mu_0 = -0.135$ in the MGCAMB parameterisation) with $\Sigma = 1$.

8 Summary

References

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System	Scale	Quantity	Value
Hydrogen atom	10^{-10} m	$\langle K \rangle / \langle V \rangle $	1/2 (13.6057 of 27.2114 eV)
20 atoms, 10 molecules (HF limit)	10^{-10} – 10^{-9} m	$T/ V $	1/2, exact for converged wavefunctions
White dwarfs	10^7 m	M_{Ch}	$1.456 M_{\odot}$ ($\mu_e = 2$)
The Sun	10^9 m	$ U /2L$	24 Myr (hence nuclear burning)
Clusters of galaxies	10^{23} m	dynamical vs lensing mass	agree to tens of percent
Simulated halos (10^{10} – $10^{15} M_{\odot}$)	10^{21} – 10^{23} m	$2K/ W $	1.02–1.17 with surface pressure
Schwarzschild black hole	r_s	$T_H S / M c^2$	1/2 (Smarr)
Cosmic horizon	10^{26} m	β_m / Ω_m	1/2, predicted; Planck $\Delta\chi^2 = +0.54$